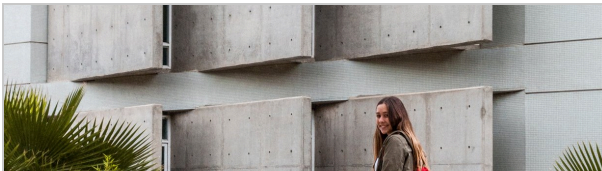


1



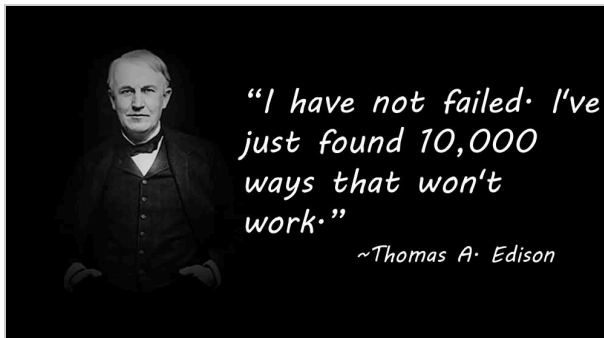
Datacommunicatie en netwerken

Bachelor in de elektromechanica – automatisering

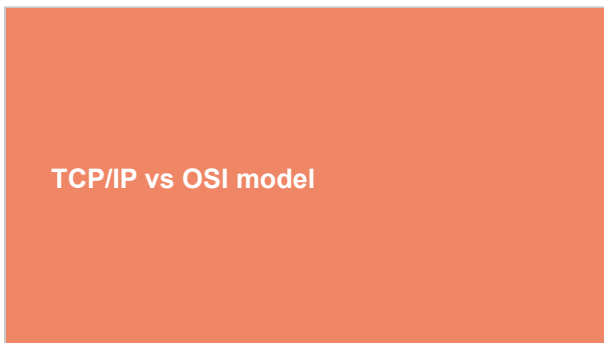
Tom Demets | Stefan Lievens

tom.demets@hogent.be | stefan.lievens@hogent.be

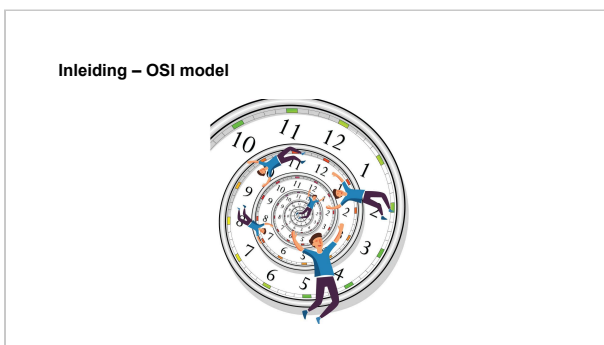
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3



4



Inleiding – OSI model

5

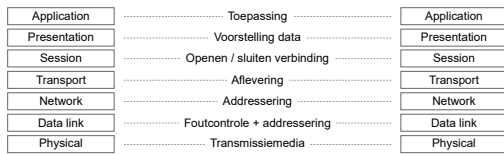
Inleiding – OSI model

Wat zou er nodig zijn qua hardware, afspraken, ... om wereldwijd computers met elkaar te laten communiceren?

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Inleiding – OSI model

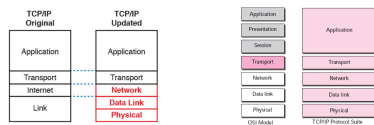
- OSI = Open Systems Interconnection (1979)
 - Set van afspraken ontwikkeld om communicatie tussen verschillende hardware mogelijk te maken => lagenmodel



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Inleiding – TCP/IP model

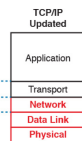
- Vereenvoudigd OSI model
 - Applicatie laag neemt Application, Presentation en Session over
- OSI in de praktijk, praktisch ieder netwerkapparaat ondersteunt TCP/IP



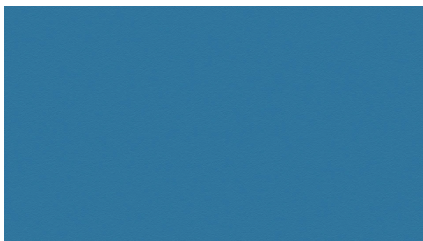
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Inleiding – TCP/IP model

- De werking van een computernetwerk begrijpen
- =
- Het TCP/IP model begrijpen
- We bespreken systematisch de verschillende lagen
- Please Do Not Take Anything (TCP/IP)*
- Please Do Not Take Something Precious Away (OSI)*



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<https://www.youtube.com/watch?v=cA9ZJdqzOoU>

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TCP/IP model / Laag 1: fysieke laag

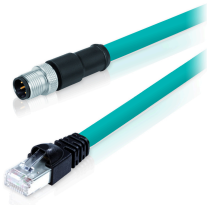
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Ethernetkabel



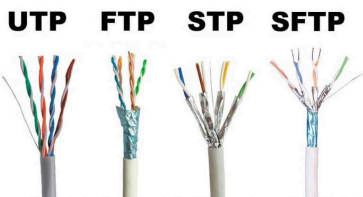
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Ethernetkabel – connectoren



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Ethernetkabel – afscherming



14

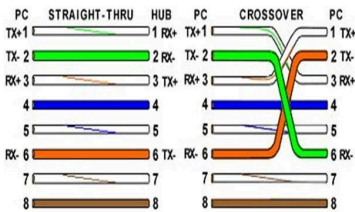
Ethernetkabel – categorie

Categorie	+/- 30m	+/- 50m	100m	Connector
cat5e				
cat6				
cat6a				
cat7				
cat8				

Hoe kan je zien als een kabel cat5e, cat6, cat6a, cat7 of cat8 is?
Wat als je verder moet dan 100m?
Wat in de industrie?
Oefening: ga op onderzoek en vul de tabel aan.

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Ethernetkabel – straight-thru en crossover



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Oefening RJ45 vs M12



M12 versus RJ45 Ethernet connection systems

Dietmar Röring, Global Key Account Manager
Phoenix Contact GmbH & Co. KG, Blomberg, Germany

Abstract

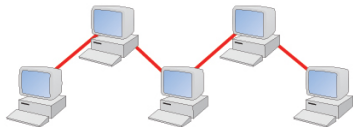
When it comes to connectivity in a facility plant environment, the people would agree upon the superiority of the M12 connector system over the standard RJ45 connector and cables, a connector originally designed to do no more than hook up a telephone extension. Industrial Ethernet calling requires connection solutions that are sturdy and reliable.

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M12 vs RJ45	3

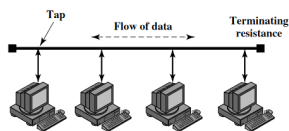
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Line topologie



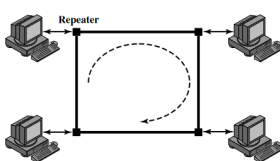
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Bus topologie



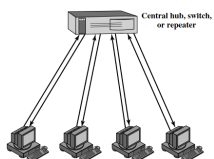
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Ring topologie



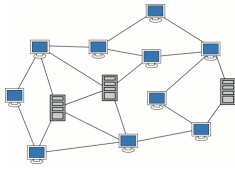
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Ster topologie



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Mesh topologie



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